

Asthm

Asthma :: type I IgE

📌 Function of alpha-1-antitrypsin in lung:

📌 *protect the lungs from neutrophils elastase & enhance alveolar exchange.*

Asthma chronic inflammatory disorder of the airways causing recurrent episodes of wheezing cough breathless chest tightness.

COPD chronic bronchitis & emphysema

Diagnosis

📌 Diagnosis is confirmed with spirometry and pulmonary function tests.

📌 Spirometry tests lung function (FEV1)= How much air can be forcefully exhaled in one second and (FVC)=After taking a deep breath, the maximum volume of air that is exhaled (how much air is exhaled).

Vaccination

📌 Annual influenza vaccine

📌 Pneumococcal polysaccharide vaccine (PPSV23)

Broncho constriction

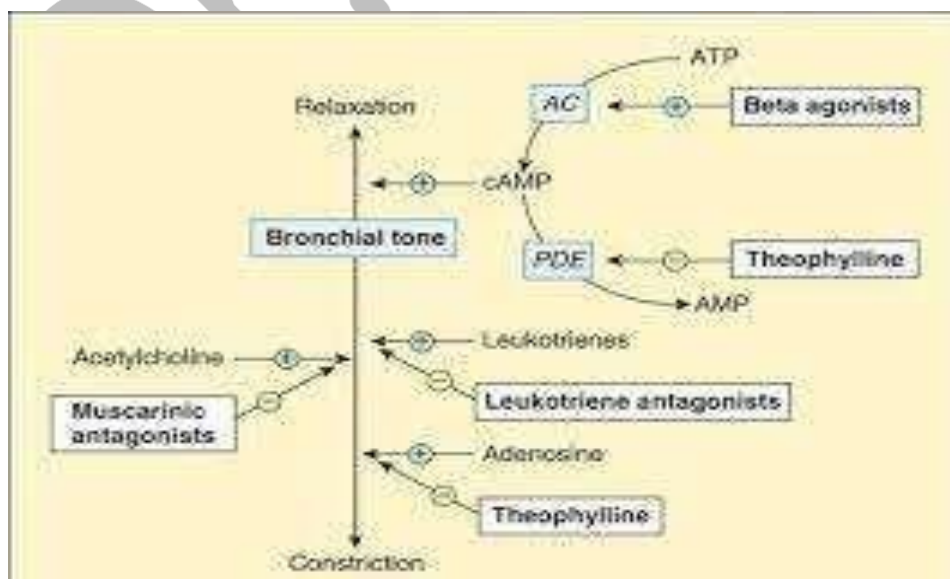
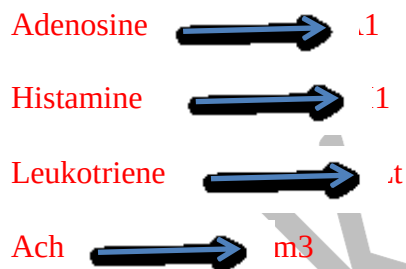
dilatation

Bronchoconstriction



dilatation

B2 agonist



Drug can cause asthma : BB, aspirin , ACE.I

Drug treat asthma :

- B2 agonist,,,
- cholinergic antagonist,, ,
- cortisone,,,
- ,Lt antagonist
- Mast cell stabilizer ,,
- methylxanthine

Dr.Kabsha

Treatment of ashtma

A. Quick relief (acute

1. SABA: (Salbutamol or albuterol) exercise induced asthma and?????
2. Systematic Corticosteroid: (Prednisone, Prednisolone, Methylprednisolone)
3. Anticholinergics: (Ipratropium) à Short acting

B. Long term control

1. Inhaled Corticosteroid: (Beclomethasone, Fluticasone, Mometasone, Budesonide)
2. LABA: (Formoterol, salmeterol)
3. Anticholinergic: (Tiotropium) à long acting.
4. Methylxanthines: (Theophylline)
5. Mast cell stabilizer (Cromolyn) (prophylactic)
6. Leukotriene receptor antagonist (montelukast, zafirlukast) aspirininduced asthma

For resistant case: anti IgE (omalizumab)

Inhaled Delivery Devices

Metered-dose inhalers (MDIs)



- ✚ Deliver a dose of aerosolized liquid medication
- ✚ Shake well in most cases
- ✚ Administration requires a **slow and deep** inhalation at the same time as pressing the canister to deliver the dose
- ✚ A spacer can be used for patients who cannot use it

Dry powder inhalers (DPIs)

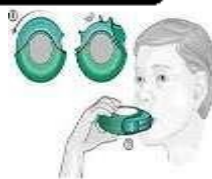


Figure 10.20 Inhaler

- ✚ Deliver a dose of fine powdered medication
- ✚ Do not shake
- ✚ Administration requires **a quick and forceful** inhalation
- ✚ Spacers cannot be used
- ✚ Pregnancy

A short-acting beta-agonist is a must. ICS are preferred controllers, typically budesonide.

B2 Agonist

➡ Short acting B2 agonist (SABA)

Ex: Salbutamol or albuterol or terbutalin

Use: (fast reliever in acute attack) **exercise induced asthma**

➡ Long acting B2 agonist (LABA) **Not used alone but in combination (ICS)**

Ex: Formoterol, salmeterol

For chronic & prevention nocturnal asthma,

Vilanterol ultra LABA ,, **indacaterol** ultra LABA in COPD

↓ S/E: tachycardia ,hyperglycemia, hypokalemia, hypomagnesimia
K, Mg

Ask about ?digoxin ,lithium , spironolactone ,ACEI
Thiazide & loop ?

➡ Anticholinergic (anti muscarinic) m3 blocker

For pt unable to tolerate B2 agonist .

Block vagal (parasympathetic)

Ex: **ipratropium** non selective muscarinic blocker **short acting** not cross BBB

Tiotropium **long acting** selective M3 blocker in COPD & chronic asthma

Atropine ???

Methylxanthines (Theophylline)

Aminophylline 80% theophylline

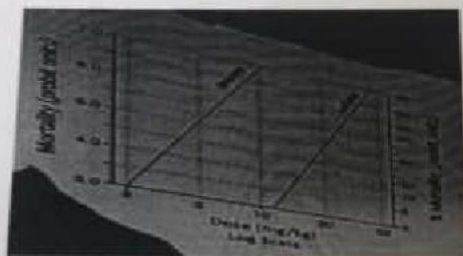
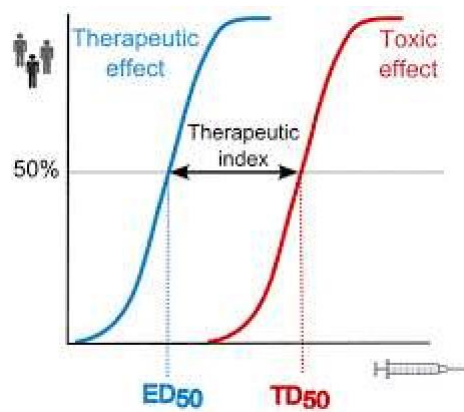
Blocks phosphodiesterase, causing an increase in cyclic adenosine monophosphate (cAMP) in

Smooth muscle >>>> bronchodilation

Cardiac muscle>>>> increase HR

Narrow therapeutic index) digoxin, Li, warfarin, tacrolimus, cyclosporine)

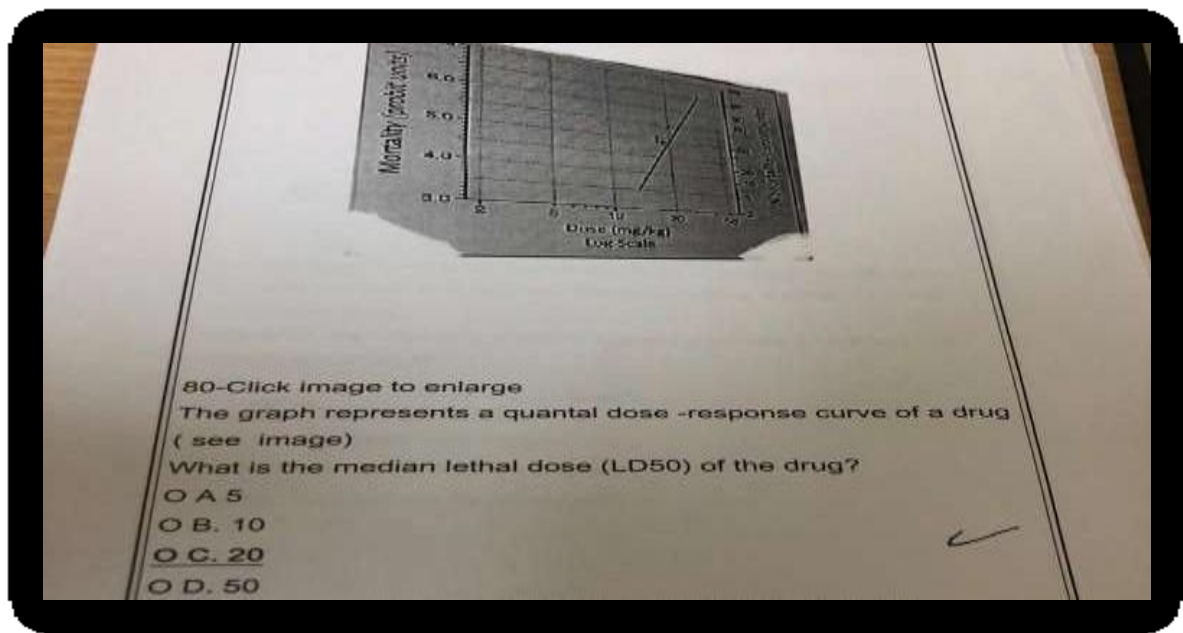
Therapeutic index: LD₅₀, or **median lethal dose**, is the dose of drug that causes death in 50% of experimental animals, and ED₅₀, or median effective dose, is the dose that produces a specified effect ("response") in 50% of the population under study.

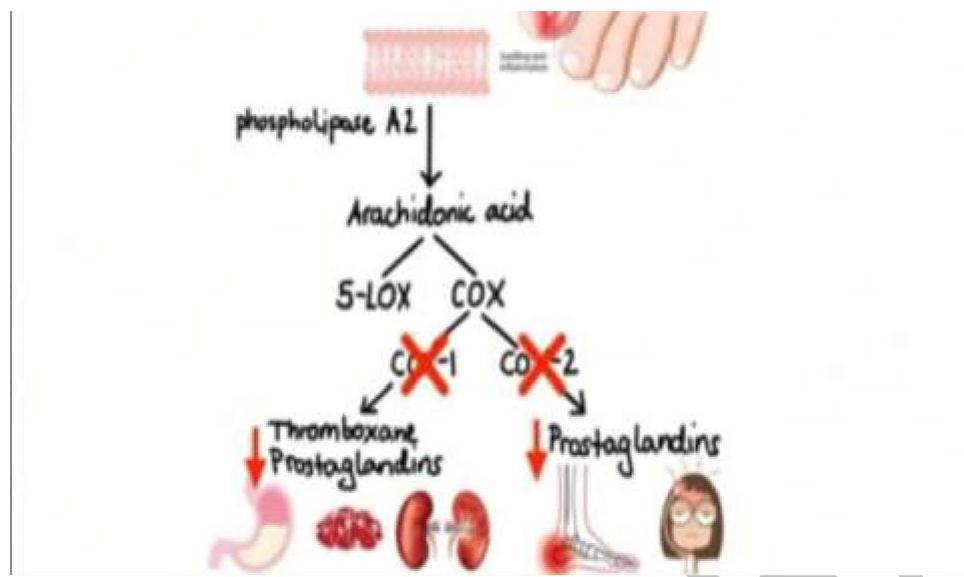


The graphs represent quantal dose-response curves for the th effect and lethality by a drug (see image).

What is the drug's therapeutic index?

- ☐ A. 0.9
- ☐ B. 1.1
- ☐ C. 4
- ☐ D. 10





Leukotriene antagonist for aspirin induced asthma

Leukotrienes receptor antagonist

for prophylaxis & ttt chronic asthma

Ex: **montelukast, zafirlukast** (hepatotoxic)

5-lipoxygenase inhibitor (chronic asthma) ex **zileutone** hepatotoxic

Corticosteroids

1. **Systematic Corticosteroid**: (Prednisone, Prednisolone, Methylprednisolone)

➡ Not use for long time

➤ Quick relief (acute cases) status asthmaticus ??

2. **Inhaled Corticosteroid (ICS)**: (Beclomethasone, Fluticasone, Mometasone, Budesonide)

➡ 1st line and **DOC** in chronic asthma, Consider Ca++ & Vit D supplements

➡ **SE**: oropharyngeal candidiasis (thrush) Wash mouth after each use OR use ciclesonide

Corticosteroids

Mineral corticosteroids: Fludrocortisone, Deoxy-corticosteroid, Aldosterone

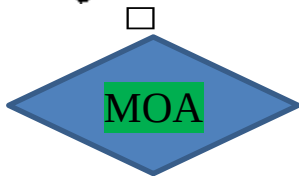
Most potent: Beta _ dexamethasone

Least potent: hydrocortisone

Max prednisone dose in asthma: 60 mg

Glucocorticoid	Equivalent dose
Cortisone	25 mg
Hydrocortisone	20 mg
Prednisone	5 mg
Prednisolone	5 mg
Methylprednisolone	4 mg
Triamcinolone	4 mg
Betamethasone	0.6 – 0.75 mg
Dexamethasone	0.75 mg

➡ Omalizumab (anti IgE)



Monoclonal antibody that inhibits IgE binding to the IgE receptor on mast cells and basophils.

for **allergic asthma** ,, **sever presistant asthma**

➡ mast cell stabilizer (cromolyn , nedocromil sodium)

prophylaxis from asthma

*Cromolyn can be used in chronic asthma in **pregnant***

For the Max. Effects take before **1-2 wks** before contact with allergen

Summary :

Exercise induced Asthma?

Aspirin induced Asthma?

Prophylaxis ?

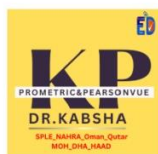
Status asthmaticus ?

Sever presistant?

Untolerate B2 agonist?

S/E of ICS ?

Classification of Asthma



Component of severity		Classification of asthma severity			
		Persistent			
		intermittent	Mild	Moderate	Severe
Impairment	Daytime symptoms	≤ 2 Days/ week	> 2 days/week but not daily	Daily	Throughout the day
	Nighttime awakening	≤ 2 x / month	3-4x/month	> 1x/week but not nightly	Often (7x/week)
	SABA rescue inhaler use	≤ 2 days /week	> 2 days/week, but not daily or > 1x/day	Daily	Several times per day
	Activity limitations	None	Minor limitation	Some limitation	Extremely limited
	Lung function FEV1 (% predicted) FEV1/FVC*	>80% Normal	>80% Normal	60-80% Reduced 5%	<60% Reduced 5%
Risk	Exacerbations requiring oral systemic steroids	0 – 1 / year	≥ 2 /year		

A 20-year-old woman presents to the clinic with an asthma exacerbation. She states that she has been using her boyfriend's albuterol inhaler on a regular basis for the past 2 years. During the past few months, she has been using the inhaler on a daily basis and sometimes at night. Which description best classifies her asthma severity?

- A. Mild intermittent.
- B. Mild persistent.
- C. Moderate persistent.
- D. Severe persistent

A 23-year-old woman has been coughing and wheezing about twice weekly, and she wakes up at night about three times per month. She has never been given a diagnosis of asthma and has not been to a physician, she says, "in years." She uses her roommate's albuterol inhaler, but having recently run out of refills, she is seeking care. Her activities are not limited by her symptoms. Spirometry today reveals FEV1 82% of predicted.

According to the NAEPP guidelines, which is the best classification of her asthma?

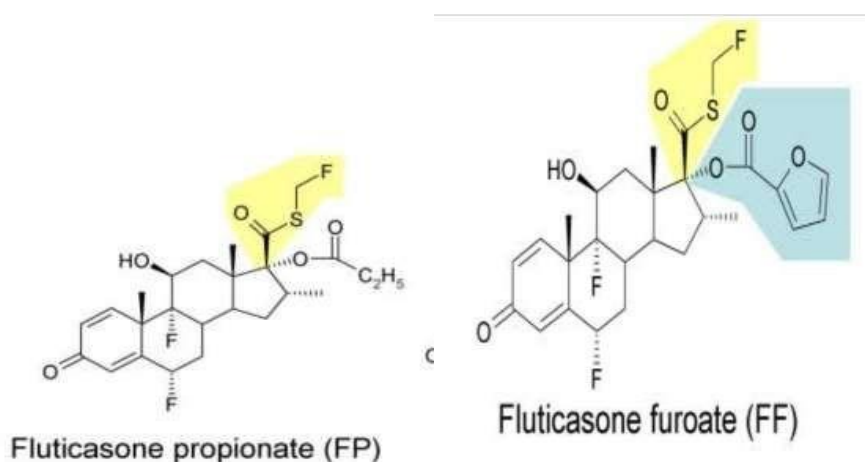
- A. Intermittent.
- B. Mild persistent.
- C. Moderate persistent.
- D. Severe persistent

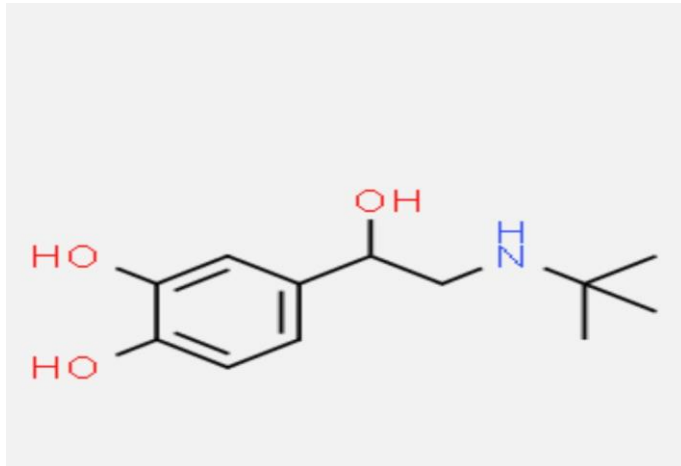
A structures of inhaled corticosteroids and the patient want a once daily inhaler, so what do you recommend according to the structure?

A-Fluticasone furate)

B-Betamethadone

C-Fluticasone propionate





Patient with respiratory depression from anesthesia, what is drug for post anesthesia respiratory?

Depression: **PicROTOXIN**

+ *Used as central nervous system stimulate, antidote*

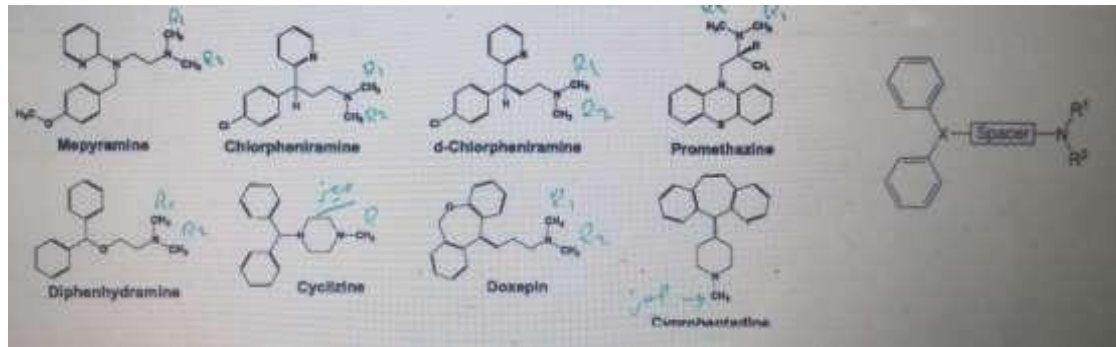
Immediate relive allergy after cleaning: **Chlorpheniramine**

GOLD SCORE COPD

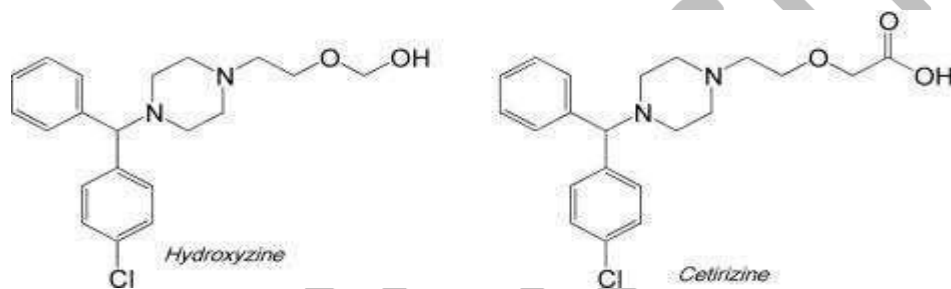
Exacerbation		one /year	2 or more / year	A or B	C or D
mMRC	1	A	one / year	bronchodilator	
mMRC	> 1	B	one / year	long bronchodilator (LABA or LAMA)	
mMRC	1	C	2 or more / year	LAMA	
mMRC	> 1	D	2 or more / year	LAMA LAMA +LABA LABA + ICS	

Anti histaminic

1st generation pass CNS cause sedation



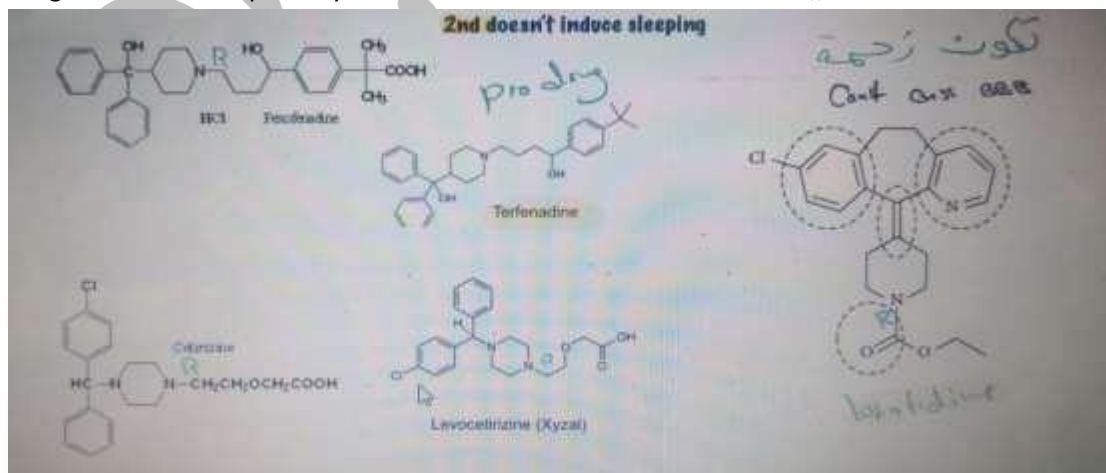
Cinazirine ,, cyproheptadine ??



(hydroxyzine)essentially acts

as a (prodrug) for the second-generation H₁ blocker, (cetirizine)

2nd generation more polar by COOH can not cross CNS less sedation ,,more selective



2nd & 3rd??? Cetirizine ,, ,, levocetirizine,, fexofenadine ,, loratidine ,,terfenadine ,, astemizole Ebastine ,, astelin .

H₃ receptor antagonist **betahistine** in treatment **Meniere disease**

SOFA Score: Septic Shock

§ Curbs65 Score = pneumonia CAP

§ Cha2ds2vasc =atrial fibrillation

§ Glascoma Score = Central nerve system

§ Child Pugh Score: liver disease

§ MELD-PLUS Score: Asses severity of Chronic Liver Disease

ABCD2 Score: Asses risk of Stroke after transient ischemic attack TIA

§ Alvarado Score: Diagnosis of Appendicitis

§ ACR Score: Rheumatoid Arthritis

§ Geneva Score: Pulmonary Embolism

§ Q RISK3 Score: Cardiovascular disease

Sofa score : septic shock

Narjeno score: Estimating the probability of adverse drug.

Doubtful: 0

Possible: 1-4

Probable: 5-8

Definite: > 8



Pain in children score:

Till 7 years ... face pain scale revised

Above 7 years Numeric rating scale

QUESTIONS

The most side effect associated with ICS inhaled corticosteroids?

Oral thrush

To prevent asthma induced by exercise give?

SABA

Side effect of albuterol ?

Tachycardia

Morphine causes respiratory depression by?

Depression of respiratory center in brain

Case of an asthmatic patient e lab data results (all of them were in normal range except for k level was lower than normal) the abnormality in test results is due to? **inhalation LABA**

Which can cause Cardiac death ?

LABA

Volume of distribution of theophylline 0.3-0.7 L/kg. Plasma protein binding: Approx 40-60%.

Non pharmacological treatment of asthma?

breathing exercise

drug used in moderate to severe asthma ?

Omalizumab

Treatment of patient with chronic wheezing and cough?

Corticosteroid

Which drug is contraindicated with asthma?

Beta 2 antagonist

Mechanism of using salbutamol inhaler?

exhale, shake the inhaler , put it between your lips then breathe slowly

Which of the following drug taken by inhalation?

Amyl nitrite or halothane

Treatment for severe allergic asthma? omalizumab used in Treatment of moderate to severe persistent asthma